

PARK FALLS, WI, USA

WMO: 727410

Lat: 45.955N

Lon: 90.424W

Elev: 1500

StdP: 13.92

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-11.3	-31.3	1.0	-14.7	-23.3	1.6	-11.1	20.8	19.0	18.8	17.6	5.8	290	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	84.8	69.1	81.7	67.3	78.9	65.7	72.1	81.1	69.8	78.5	67.9	76.2	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	112.4	78.1	66.7	103.7	75.1	64.8	97.1	73.0	36.7	81.0	34.8	78.5	33.0	76.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.4	16.4	13.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-21.0	75.6	7.6	2.1	-26.4	77.2	-30.9	78.4	-35.2	79.6	-40.7	81.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	13.3	16.0	27.3	39.6	55.7	64.4	66.3	64.6	55.4	42.2	28.6	18.4
	DBStd	21.38	12.46	12.74	11.75	9.10	9.23	7.49	5.55	5.93	7.84	8.05	9.21	11.31
	HDD50	5104	1137	951	706	333	49	3	0	0	35	266	644	980
	HDD65	8986	1602	1371	1167	763	310	99	50	82	298	707	1092	1445
	CDD50	1866	0	0	3	20	226	434	506	452	199	24	2	0
	CDD65	273	0	0	0	0	22	80	92	68	11	0	0	0
	CDH74	2092	0	0	0	5	251	691	611	482	50	2	0	0
	CDH80	437	0	0	0	0	37	181	123	94	2	0	0	0
	WSAvg	6.5	7.1	6.9	6.9	7.3	6.4	5.8	5.7	5.5	6.1	6.5	7.3	6.7
	PrecAvg	33.1	1.1	0.9	1.8	2.6	3.5	4.4	4.5	3.7	4.1	3.4	1.8	1.2
(15) Precipitation	PrecMax	43.7	2.6	2.0	3.6	5.8	6.2	9.0	9.1	9.2	8.3	7.7	4.4	2.2
	PrecMin	24.6	0.4	0.1	0.3	0.8	2.1	1.2	1.3	1.3	1.0	0.9	0.3	0.4
	PrecStd	6.1	0.6	0.5	0.9	1.3	1.2	1.9	2.0	1.9	2.1	1.8	1.0	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	44.5	65.3	73.6	83.4	88.0	88.1	87.1	78.6	68.9	59.1	40.0
		MCWB	34.0	38.1	52.1	57.3	61.8	69.8	73.2	71.5	64.7	57.5	51.7	34.7
	2%	DB	35.9	39.8	55.7	65.2	80.0	84.6	83.2	83.4	75.3	64.2	51.1	36.0
		MCWB	33.0	34.1	45.5	51.2	62.2	67.8	70.6	69.7	63.6	53.4	46.3	32.2
	5%	DB	32.8	36.0	48.1	60.2	76.5	81.3	80.0	79.7	72.2	60.2	46.7	33.7
		MCWB	30.5	31.6	39.7	49.7	60.1	67.1	67.8	67.2	61.9	51.4	41.7	31.2
	10%	DB	30.2	32.3	42.7	55.4	72.2	77.8	77.3	76.1	68.5	56.1	42.0	31.5
		MCWB	28.4	29.5	36.3	45.4	58.5	64.3	66.0	65.1	60.9	48.3	37.2	29.7
	0.4%	WB	35.4	39.4	53.1	58.9	67.6	73.1	75.6	74.4	68.6	59.1	54.6	35.2
		MCDB	37.5	44.3	63.3	68.3	78.1	84.1	83.8	84.5	74.1	64.9	58.2	38.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.9	34.3	45.8	54.6	65.2	70.7	72.4	71.3	66.2	55.8	47.7	32.9
		MCDB	35.7	39.1	54.1	62.7	74.9	81.4	80.7	79.6	71.7	62.2	50.7	35.3
	5%	WB	30.7	32.0	40.6	49.9	63.2	68.3	70.0	68.9	64.4	52.1	41.7	31.3
		MCDB	32.8	35.8	47.3	59.8	71.9	78.5	77.8	77.3	70.0	58.8	46.2	33.3
	10%	WB	28.4	29.7	37.1	45.8	59.9	66.2	67.9	66.8	61.7	48.4	37.3	29.6
		MCDB	30.2	32.5	41.9	54.5	69.5	75.3	75.1	74.6	67.5	55.1	41.6	31.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	17.9	17.8	18.6	20.4	18.5	16.6	17.8	17.1	16.1	12.0	13.1
		MCDBR	16.8	18.6	25.4	26.2	25.8	21.9	19.8	21.6	21.1	24.2	17.9	12.1
		MCWBR	15.0	14.9	16.5	15.9	12.9	11.5	10.8	11.2	12.0	15.9	14.5	10.8
	5% WB	MCDBR	16.2	18.4	23.2	24.4	21.6	19.8	17.4	18.8	17.7	22.2	17.4	11.6
		MCWBR	14.8	15.2	15.6	15.6	13.1	11.7	11.0	11.1	12.3	16.6	15.2	10.8
(40) Clear-Sky Solar Irradiance	taub		0.239	0.263	0.292	0.332	0.359	0.359	0.361	0.360	0.340	0.304	0.278	0.250
	taud		2.404	2.334	2.401	2.437	2.399	2.420	2.435	2.448	2.499	2.584	2.544	2.441
	Ebn at Noon		283	293	299	294	288	287	285	282	279	277	266	266
	Edh at Noon		22	29	32	34	36	36	35	33	29	23	20	19
(44) All-Sky Solar Radiation	RadAvg		471	816	1187	1437	1687	1845	1928	1613	1241	768	455	327
	RadStd		50	72	108	148	127	138	80	109	76	88	51	37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page